DocStock

Reminder a day keeps Doc at bay

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Team Pearl



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Isaiah Gamble is a senior at ODU majoring in Computer Science with a minor in Mathematics. He is a Coast Guard veteran and has "Experience Every Natural Disaster" on his bucket list. He plans on pursuing the arts after completion of his degree.

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Julia Hairston

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Anson Cheng

Team Pearl



Ab Aljazaeri

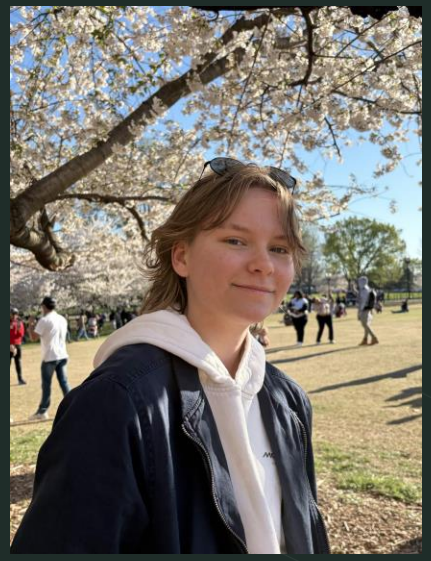
Abdelrahman “Ab” Aljazaeri is a senior at Old Dominion University majoring in Computer Science with a minor in Mechanical Engineering Technology. He likes to tinker with cars and dabbles in network security. He plans to keep working where software meets hardware after graduation.

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Toni Sallaku

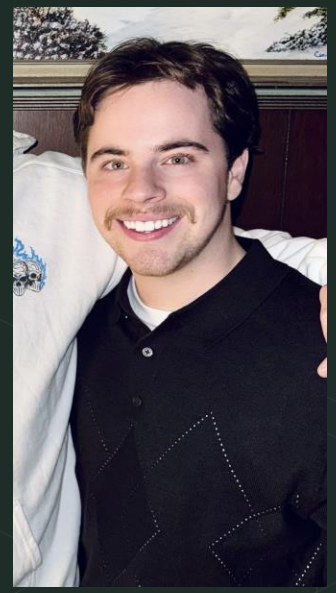
Team Pearl



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Matthew Richards

Problem Statement

Many people who take medications regularly have difficulty managing their medication schedules, remembering doses, tracking whether medications were taken, and knowing if their supply is running low. These difficulties can lead to missed, delayed, or repeating doses of medicine already taken.

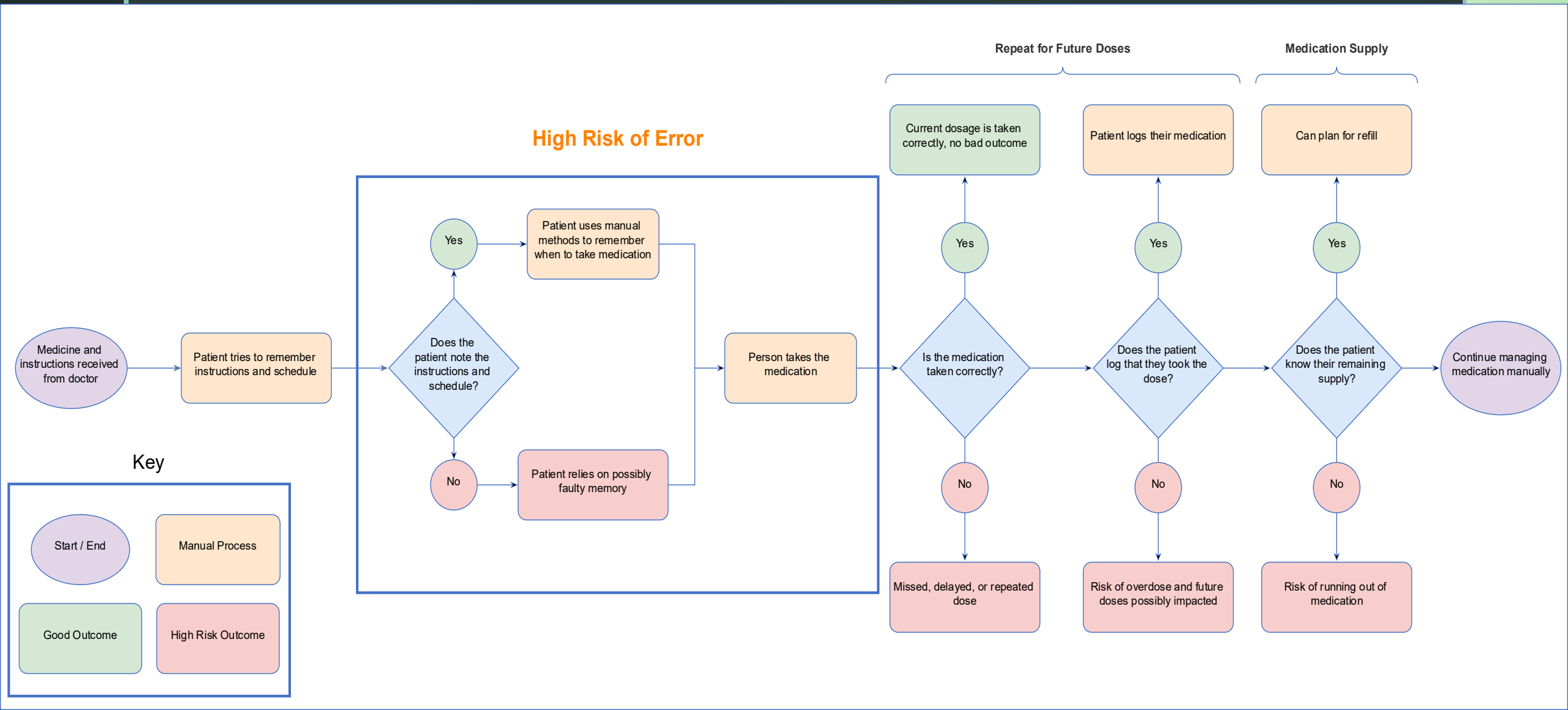
Problem Characteristics

According to the Centers for Disease Control and Prevention (CDC), about one in five new prescriptions in the United States are never filled.

Approximately 50% of filled prescriptions are taken incorrectly with regards to timing, dosage, frequency, and duration.

Direct health care costs associated with nonadherence have grown to approximately $100-$300 billion of U.S. health care dollars spent annually. [1]

Current Process Flow



Elevator Pitch

DocStock is a web and mobile app that helps people and their care givers manage prescriptions by scheduling doses, screening for drug interactions, and tracking supply, with the goal of increased medication adherence.

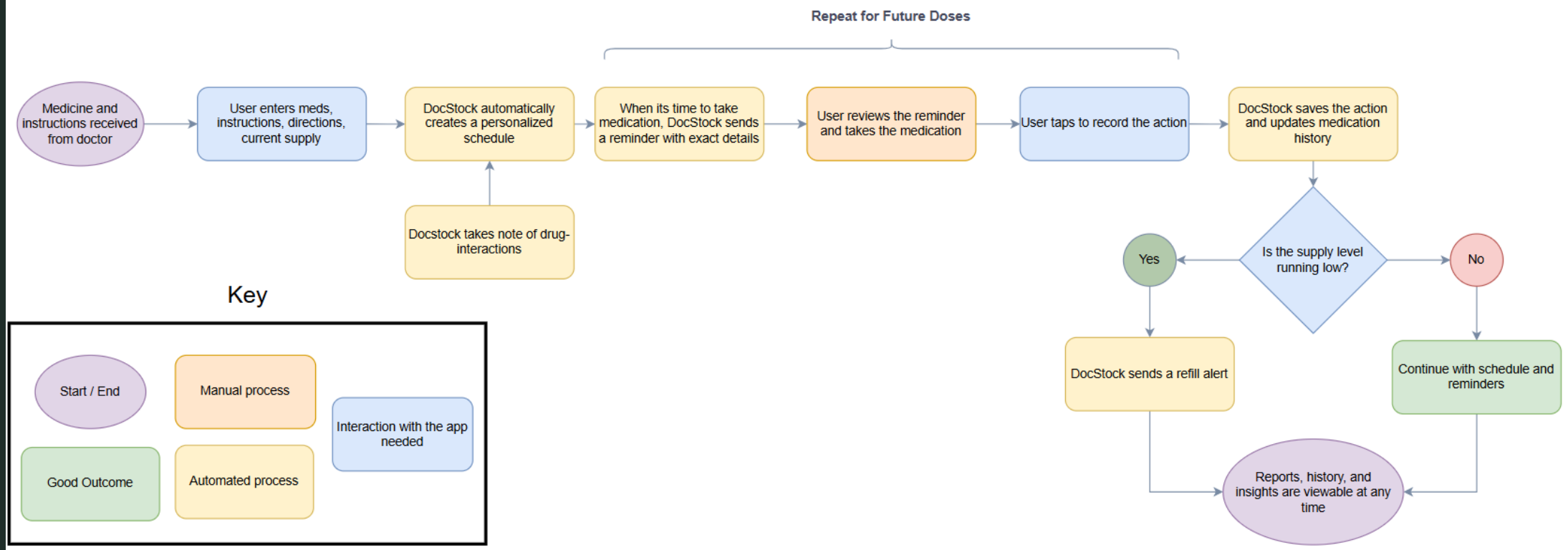
Solution Statement

DocStock will be a smartphone and web application that helps people manage their prescribed medication schedules. Users enter their medications, dosing instructions, and current supply, and DocStock organizes everything into a single, easy-to-follow schedule.

When a new medication is added, DocStock checks it against the user's active medication list and warns the user of known drug interactions. The app sends reminders when doses are due, lets users record each dose taken, tracks remaining supply, and alerts the user before a refill is needed.

Adherence history is compiled into a report that can be exported for a healthcare provider. Caregivers can be granted access, and multiple profiles can be managed under one account.

Solution Process Flow



The number of manual processes is reduced to one compared to the current process flow.

What it Will Do

- Centralize every prescription in one dashboard reachable from any device.
- Schedule doses and send reminders so doses aren’t missed, mistimed, or doubled
- Alert users before refills run out to prevent gaps in treatment
- Warn of drug interactions by checking each newly added medication against the user's active list
- Track adherence and export a provider-ready report to bring to appointments
- Share access with caregivers and support multiple profiles under one account

What it Will Not Do

- It will not automatically order refills for low medications.
- It will not give medical advice or replace healthcare professionals. Interaction warnings are informational only; users are directed to confirm with their provider or pharmacist.
- It will not log symptoms, mood, or health metrics (blood pressure, glucose, etc.)
- It will not integrate directly with pharmacies to pull prescription automatically.

Competition Matrix

Feature	DocStock	Medisafe [2], [3]	MyTherapy [4]	Round Health [5]
Dose reminder & scheduling	✓	✓	✓	✓
Prescription refill alert	✓	✓	✓	✓
Drug interaction warning	✓	✓	-	-
Caregiver mode	✓	-	-	-
Exportable adherence report	✓	Partial	✓	-
Multiple user profiles	✓	✓	✓	✓
Web-based (browser and phone app)	✓	-	-	-
Free / no paywall	✓	Partial	✓	Partial

Development Tools

- IDE: Visual Studio Code. Writing and editing code.
- Web Fronted: HTML, CSS, JavaScript. Website interface.
- Mobile Frontend: React Native. Mobile application.
- Backend: Node.js & Express. Handle server-side logic and API requests.
- Database: MySQL. Store users, medications, schedules, history, etc.
- Testing: React Native Testing Library. Test frontend/backend functionality.
- Continuous Integration & Continuous Deployment: GitHub & GitHub Actions. Automated testing and version control

Major Functional Components

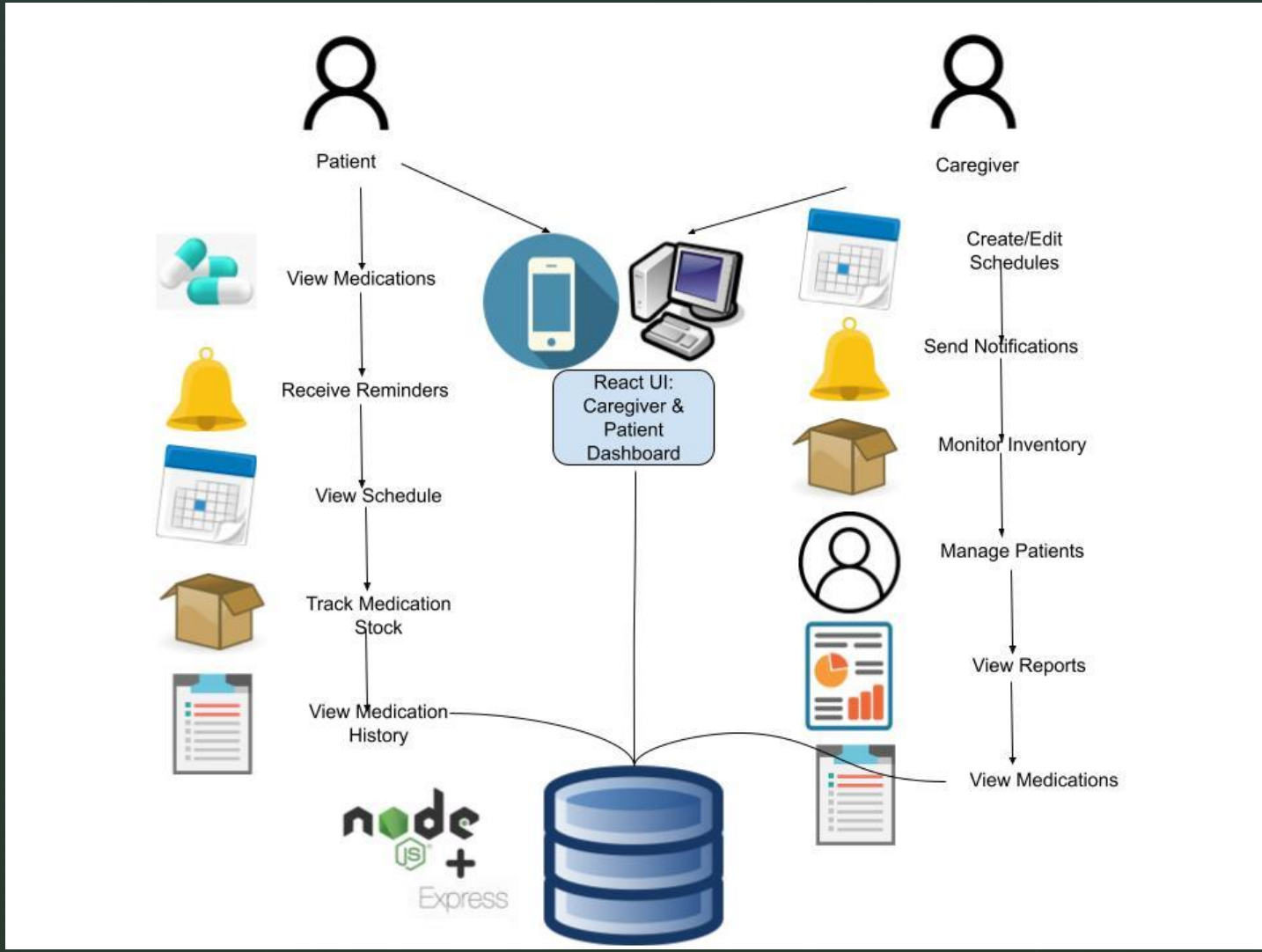
Web Application: HTML, CSS, JavaScript
- HTML for the website's structure
- CSS for styling
- JavaScript for interactivity

Mobile Application: React Native
- React Native will be used for both iOS and Android development

Backend: Node.js, Express.js
- Node.js will be the server-side environment
- Express.js will be for backend and APIs

Database: MySQL
- MySQL will store and manage user information

Major Functional Components Diagram



Customer and End User Risks

Risk	Probability	Impact	Mitigation
User forgets to enter or update their medications	3	5	The app will use a simple set-up process with clear prompted instructions for adding and updating a medication. There will be a clearly visible button to add medication.
User ignores or misses medication reminders	2	5	Provide regular notifications and an easy way to record having taken a medication.
The app is difficult for some users to navigate	4	4	Large text, obvious buttons, simple navigation. User testing can identify problems before actual release.
User enters incorrect medication info	3	5	The app will require confirmation of medication details. It will clearly state that users should follow instructions provided to them by their healthcare providers.

Technical Risks

Risk	Probability	Impact	Mitigation
Delayed or failed notifications	2	5	Test notifications across supported devices. Provide in-app schedule for users to manually track. Alert users they must turn on app notifications to get them.
Smartphone compatibility problems	3	5	Test the application on multiple supported devices and OS versions via emulators.

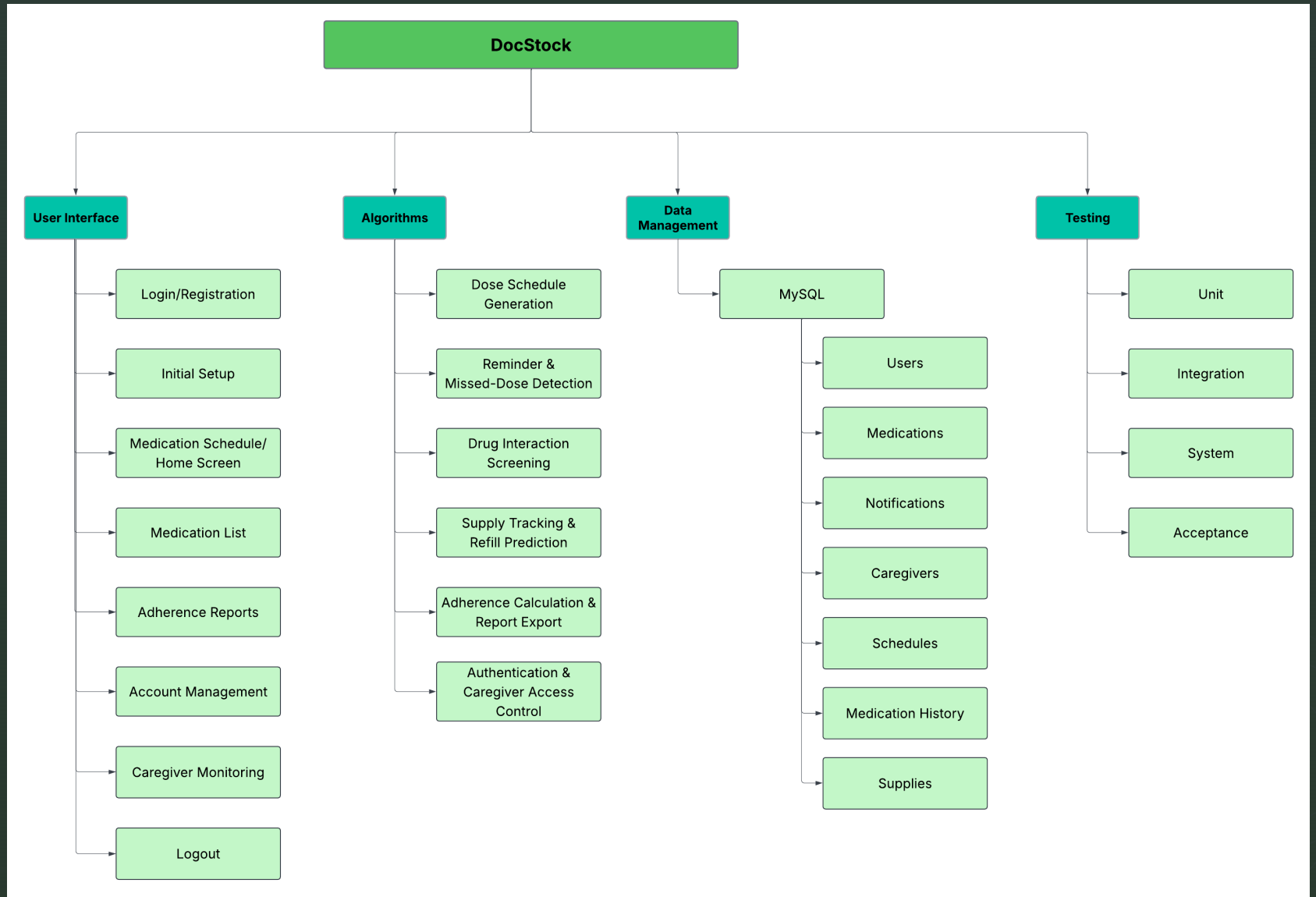
Security Risks

Risk	Probability	Impact	Mitigation
An unauthorized user gains access to medication information	2	4	App will be password protected, server side information will be encryption protected
An unauthorized user gains access to user's username and password	2	4	App will provide users with an optional two-step security verification to add an additional layer of account security [6]
User forgets username or password	4	2	App will allow users the ability to change username and password

Legal Risks

Risk	Probability	Impact	Mitigation
User claims the app caused harm because they relied on a reminder.	3	5	Communicate the app's limitations and advise users to verify medication instructions with their healthcare provider.
App unintentionally infringes on copyright or trademarks	2	3	App will use original or licensed materials with proper credit.

Work Breakdown Structure



Algorithms – Overview

Each algorithm, the feature it powers, and how much of it the prototype will build.

Algorithm	Powers	Prototype	Data (MySQL tables)
1. Dose Schedule Generation	Add medication, Today schedule	**FULL**	`schedules`, `medication_history`
2. Reminder & Missed-Dose Detection	Reminders, dose status	**FULL** – caregiver alert is real world only	`notifications`, `medication_history`
3. Drug Interaction Screening	Add-medication warning	**SIMULATED** – seeded interaction table	`medications`, `drug_interactions*`
4. Supply Tracking & Refill Prediction	Supply / refill screen	**FULL**	`supplies`, `schedules`
5. Adherence Calculation & Report Export	History / report screen	**FULL** – PDF and CSV	`medication_history`
6. Authentication & Access Control	Login, every API request	**PARTIAL** – no 2FA or caregiver access	`users`, `caregiver_access`

FULL built and demoed **SIMULATED** built with stand-in data **PARTIAL** core built, rest is real world only

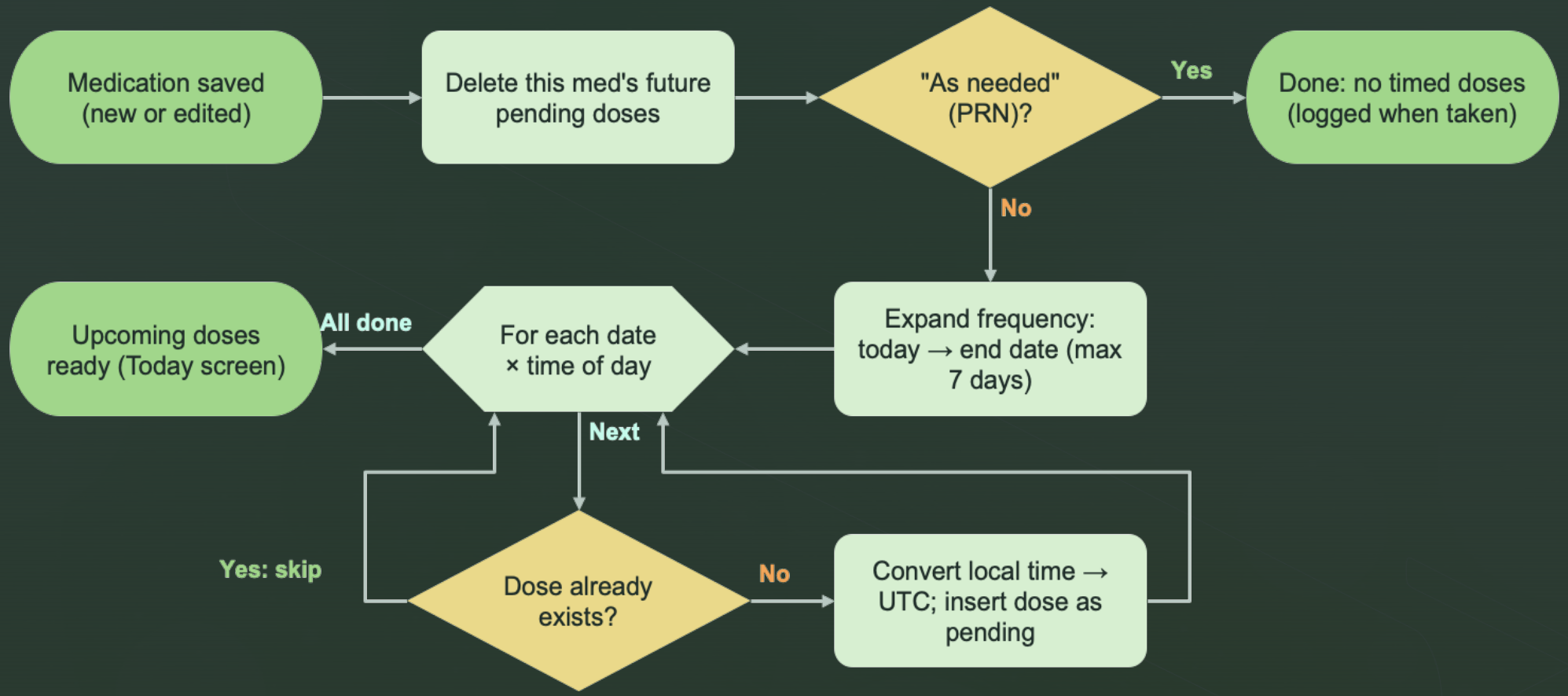
* drug_interactions is a proposed addition to the Database Schema slide.

Algorithms – Dose Schedule Generation

Turns a prescription's instructions into specific timed doses.

- Input: Medication, dose amount, frequency (e.g., 2x daily, every 8hrs, certain weekdays), start/end date, time zone
- Steps: Expand the frequency rule into dose time over the date range. Store times in UTC and display them in local time. Skip "as needed" meds. Regenerate future dose when a prescription is edited.
- Output: List of upcoming dose events per profile

Flow – Dose Schedule Generation



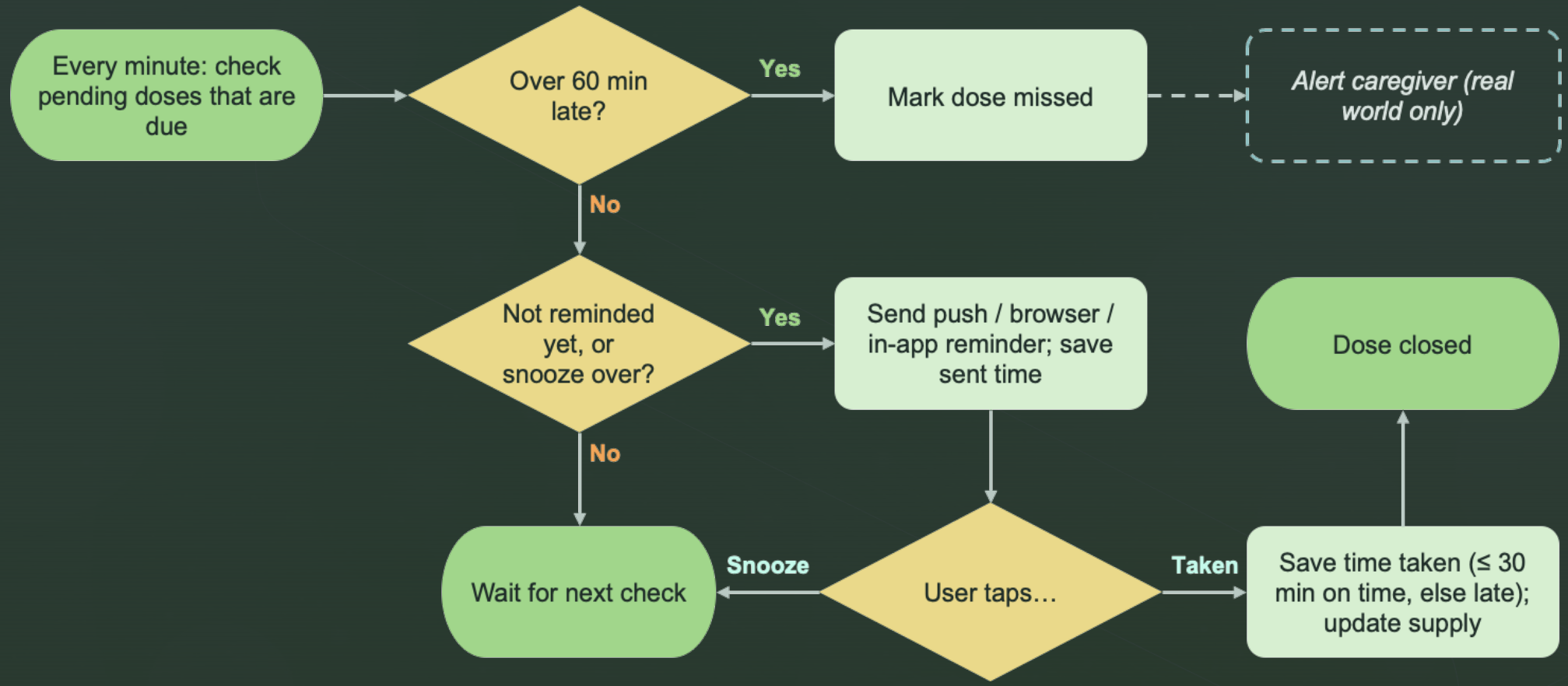
Algorithms – Reminder & Missed-Dose Detection

Decides when to notify the user and when a dose is missed

- Steps: A background job checks for doses coming due and sends a notification (push on mobile, browser notification on web, in-app as backup). Snooze reschedules the reminder, and each dose is never sent twice. If a dose isn't logged within a grace window (e.g., 60 min), it's marked missed and the caregiver can optionally be notified.
- Output: notification sent, dose status (pending/ taken/ late/ missed)

[11] Push API

Flow – Reminders & Missed Doses



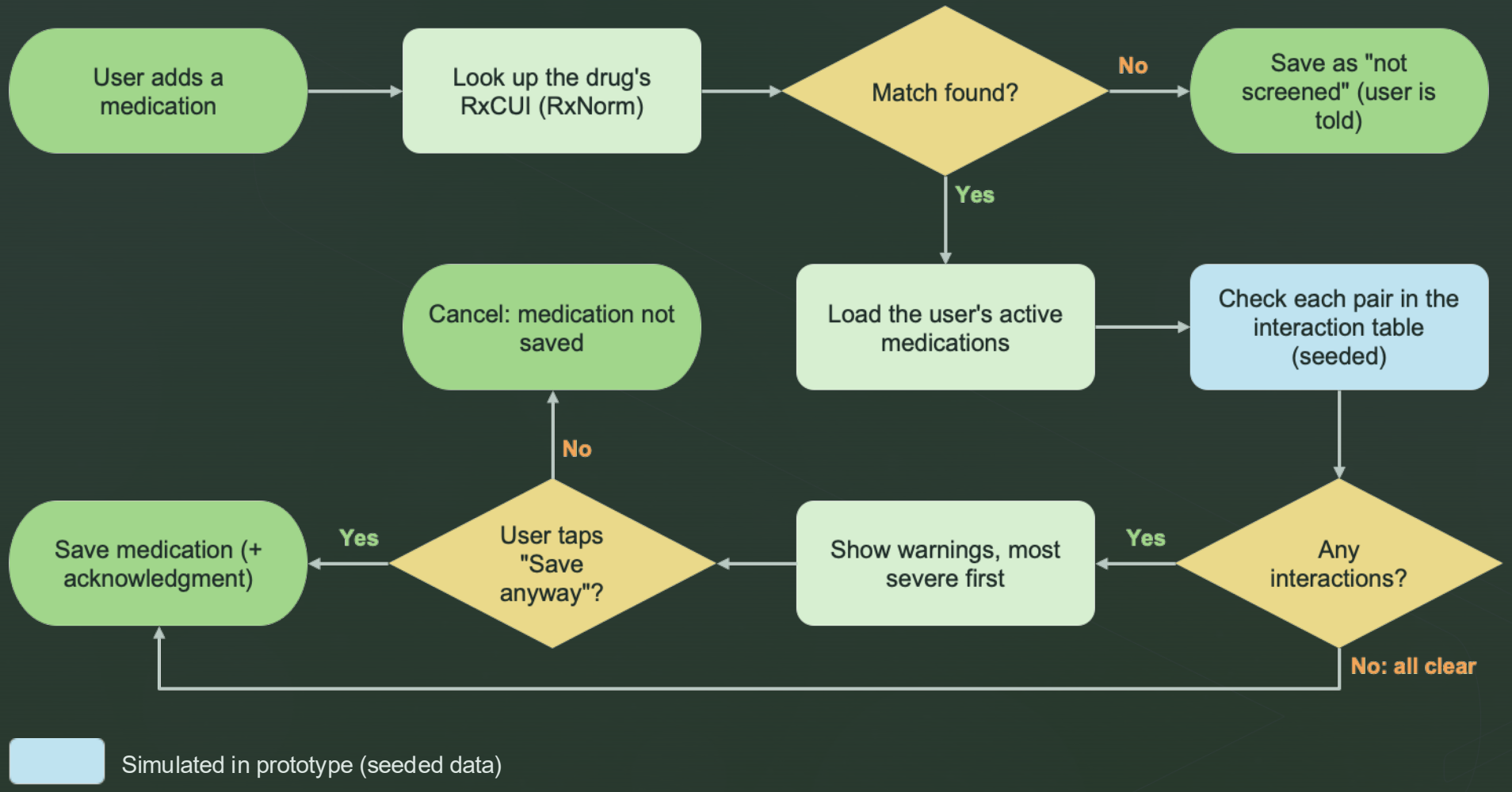
Algorithms – Drug Interaction Screening

Runs whenever a medication is added.

- Steps: Convert the drug name to a standard ID (RxNorm RxCUI). Compare the new drug against every active med in that profile, one pair at a time. Look up each pair in an interaction table. Show the severity and a plain-language warning, which the user must acknowledge.
- Output: Warning or all-clear (Informational only, not medical advice)

[7] RxCheck, [8] RxNorm API

Flow – Drug Interaction Screening

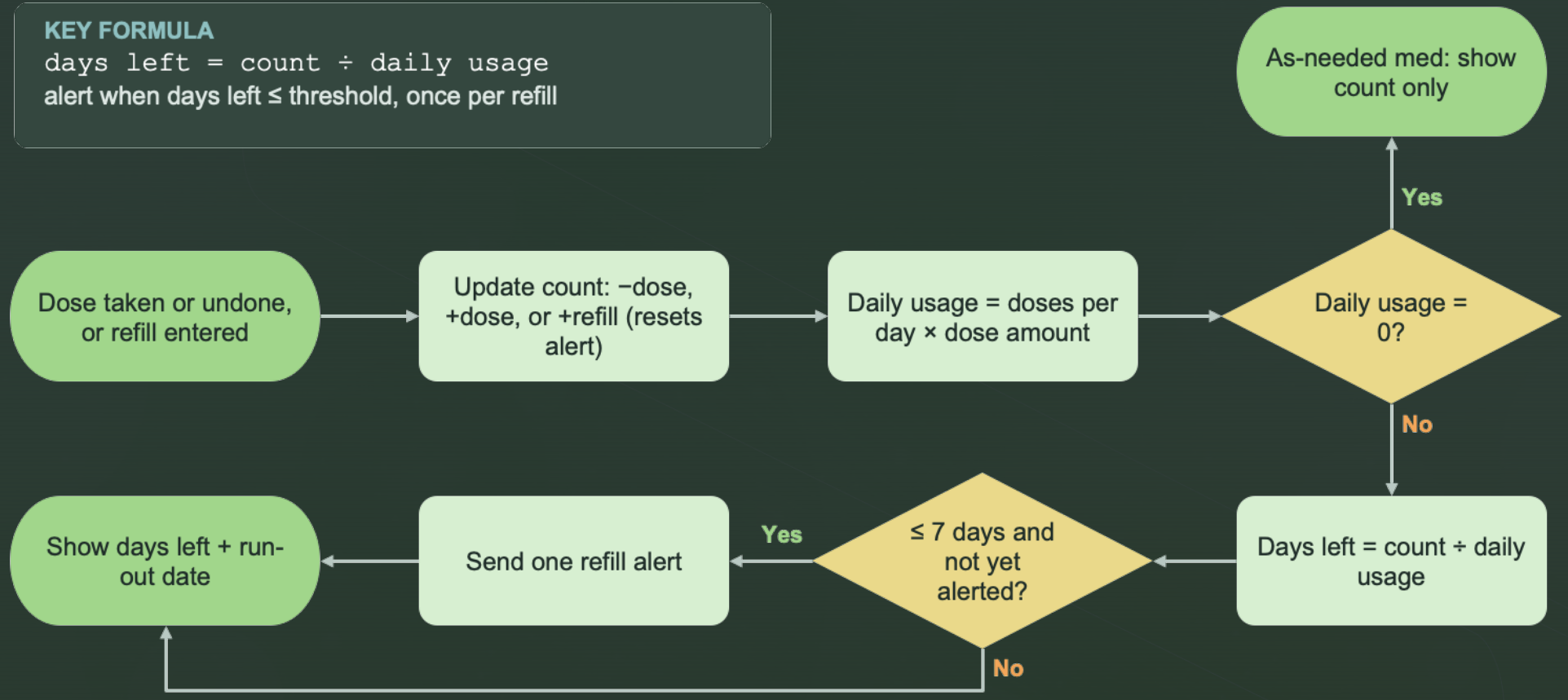


Algorithms – Supply Tracking & Refill Prediction

- Steps: Each logged dose subtracts the dose amount from supply, and undo adds it back. Daily usage comes from the schedule, and days remaining = supply / daily usage. If days remaining falls at or below a threshold (e.g., 7 days), send one refill alert. Reset when the user enters a refill.
- Output: Days of supply left, refill alert.

Flow – Supply & Refill Prediction

KEY FORMULA
`days left = count ÷ daily usage`
alert when days left ≤ threshold, once per refill

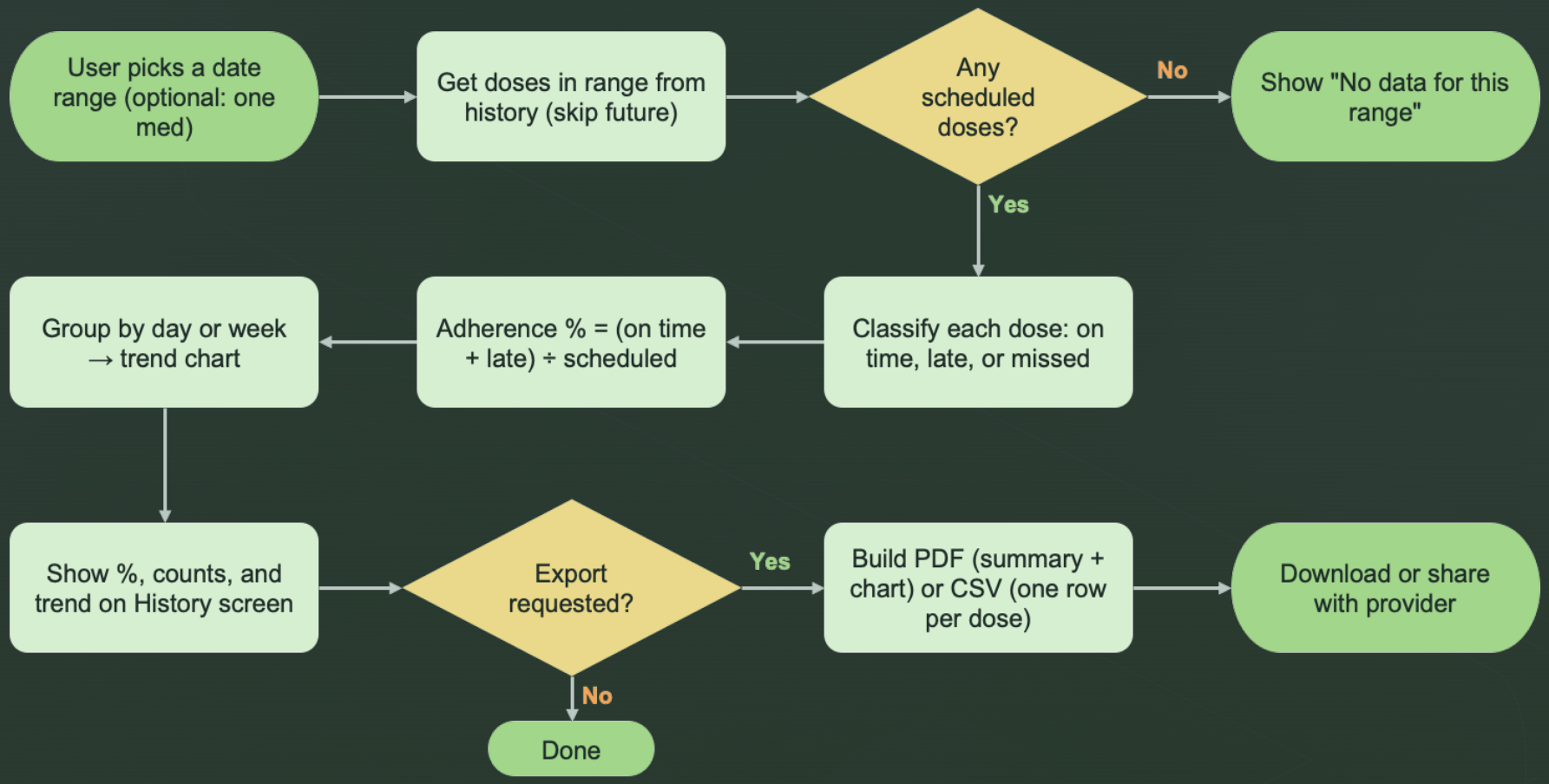


Algorithms – Adherence Calculation & Report Export

- Steps: For a chosen date range, adherence %= doses taken / doses scheduled, per medication and overall. Counts dose as on-time, late, or missed, and group them by day or week to show a trend.
- Output: Exportable PDF/CSV report for the provider

[1] CDC Adherence

Flow – Adherence & Report Export

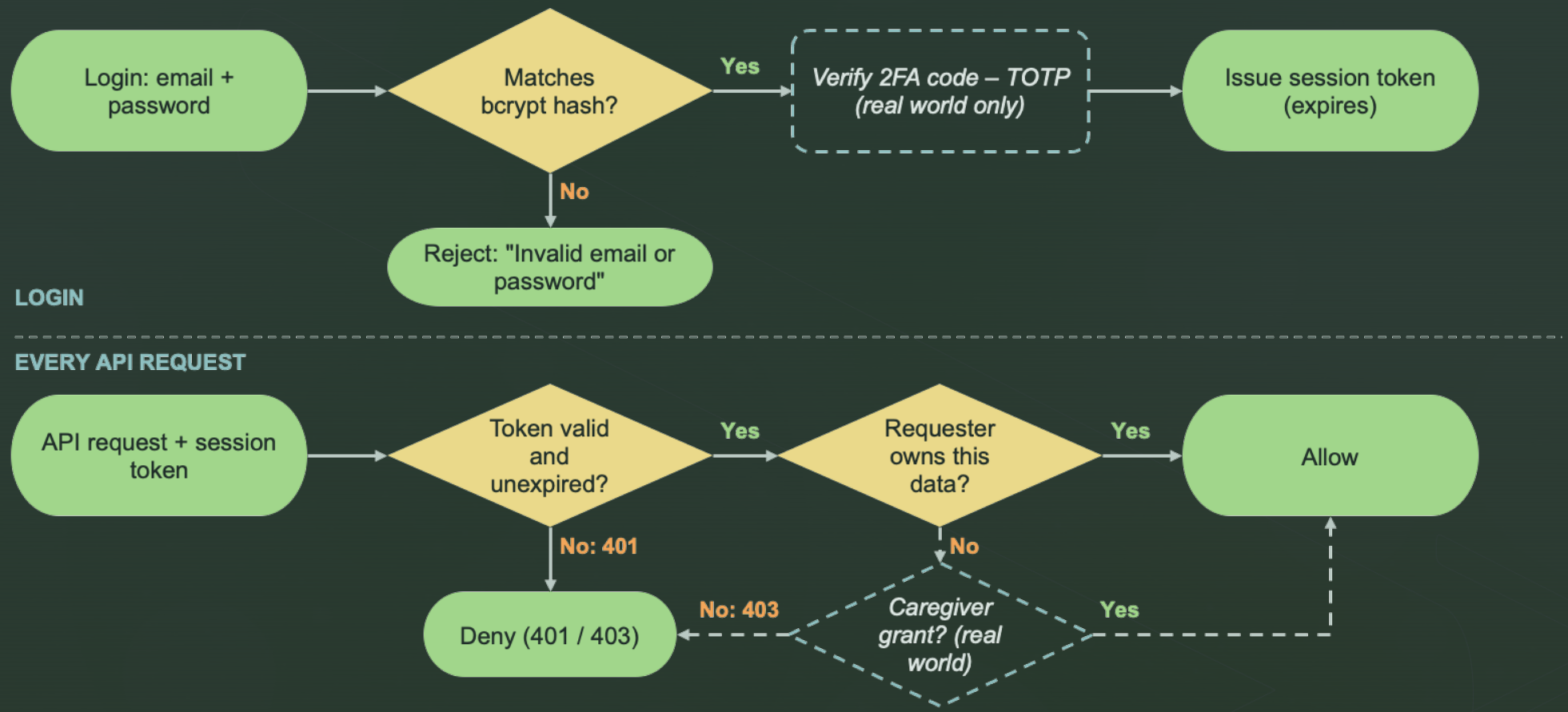


Algorithms – Authentication & Caregiver Access Control

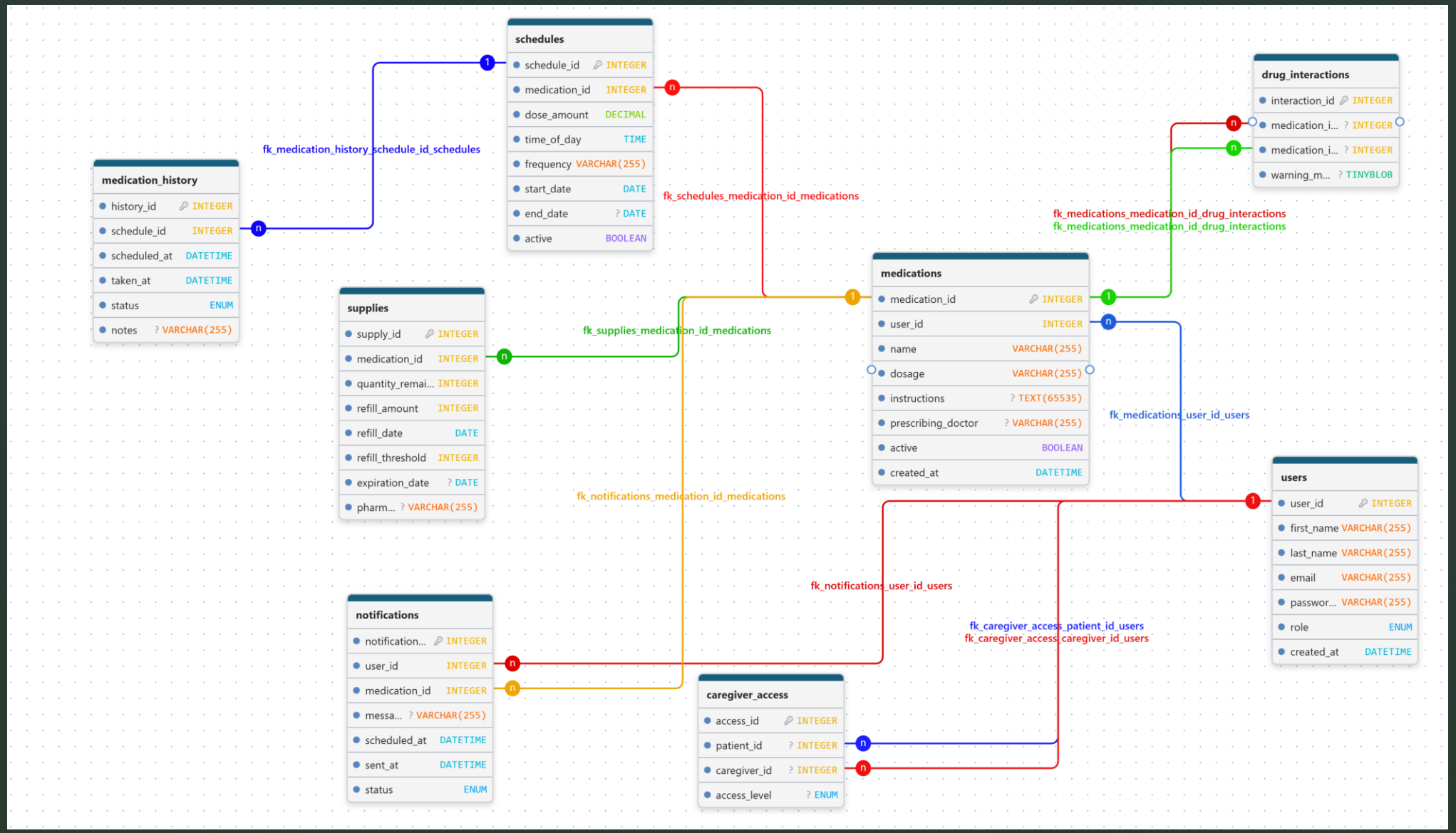
- Steps: Hash passwords (bcrypt) and support optional two-step verification. The owner sends an expiring invite link that gives a caregiver view or edit access to specific profiles. Every request checks if the user is the owner or an authorized caregiver with the right permission. The owner can revoke access anytime.
- Output: Allow or deny per request

[9] bcrypt, [10] TOTP Access

Flow – Authentication & Access



Database Schema



Real World Product vs. Prototype

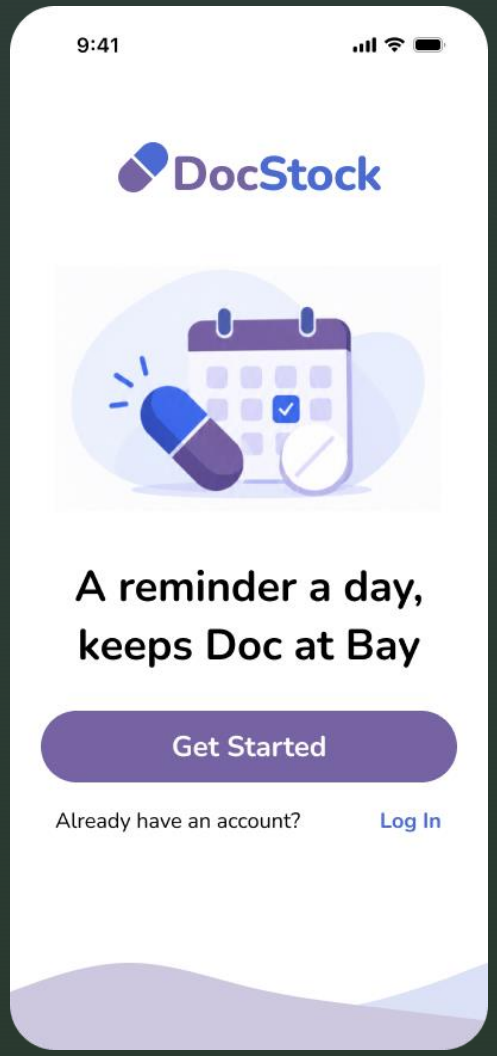
We want the prototype to be as close to the real thing as it can get. It is pivotal to have all the possible functionality down pat to display our current and subsequent ideas.

	Login with Password Protection	Initial Setup	Medication Schedule Management	Adherence Reminders and Reports
Real World Product	✓	✓	✓	✓
Prototype	✓	✓	✓	✓

Real World Product vs. Prototype

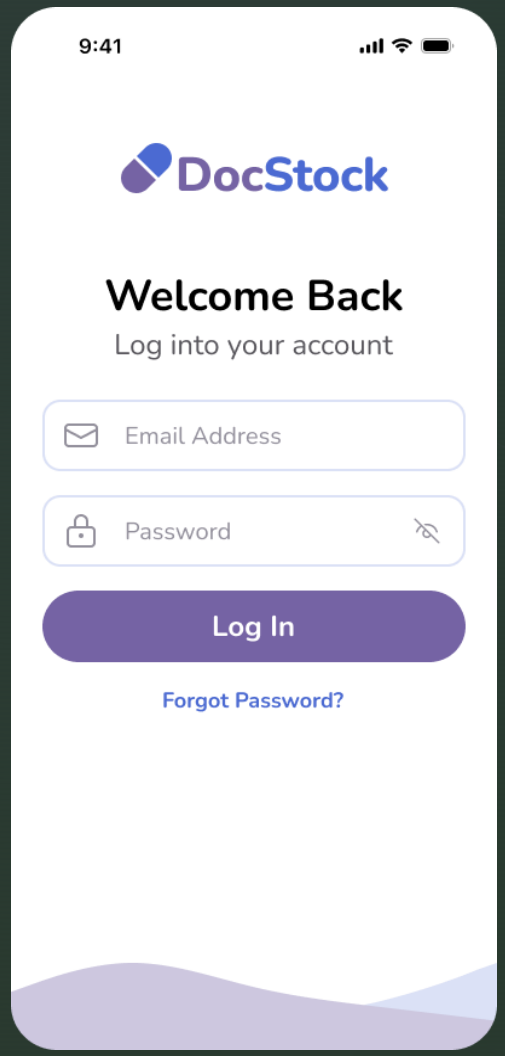
	Refill Reminders	App and Website Support	Adverse Drug Interaction Checking	Caregiver(s) Support
Real World Product	✓	✓	✓	✓
Prototype	✓	✓	X	X

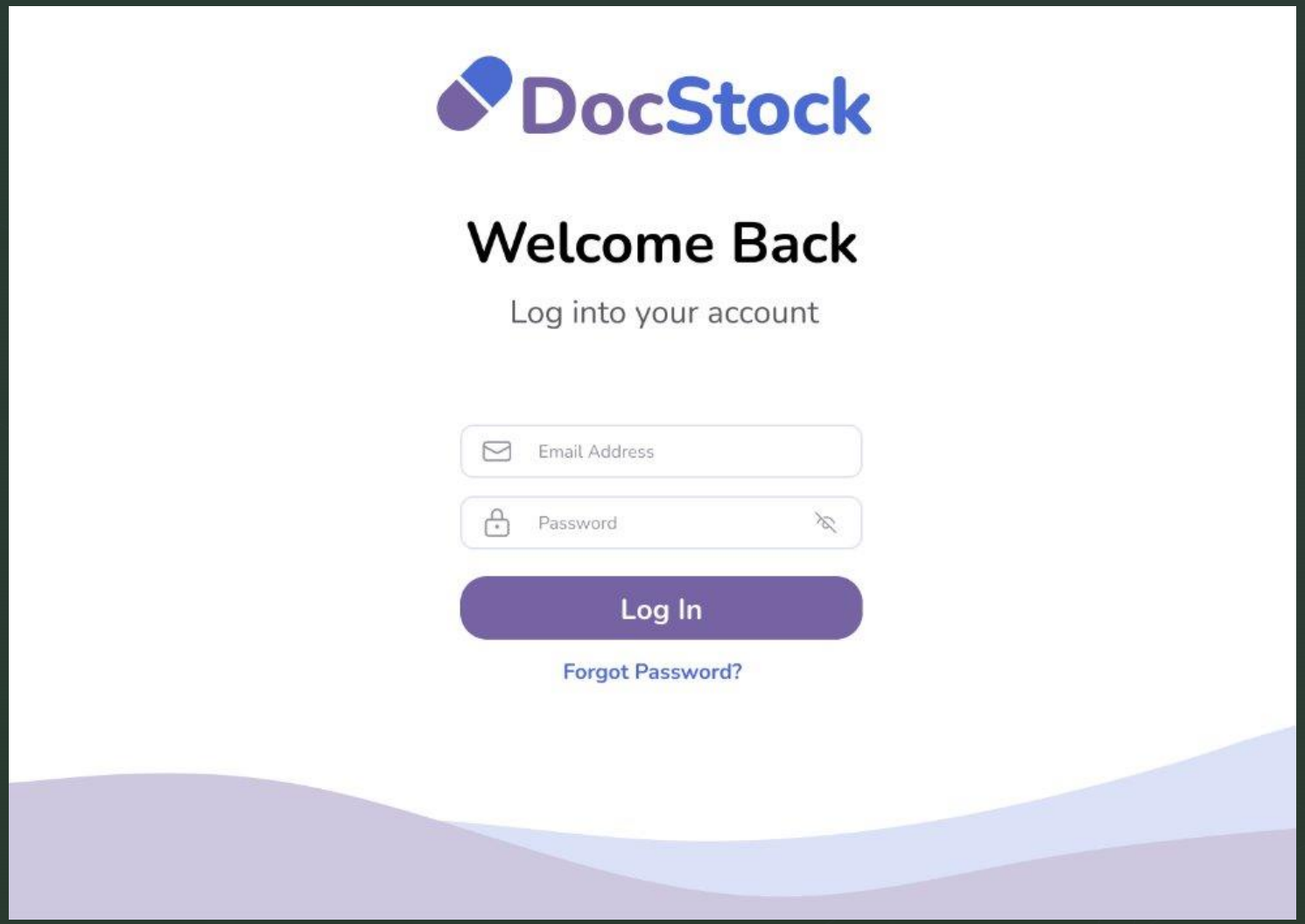
GUI Mockup: Initial Screen



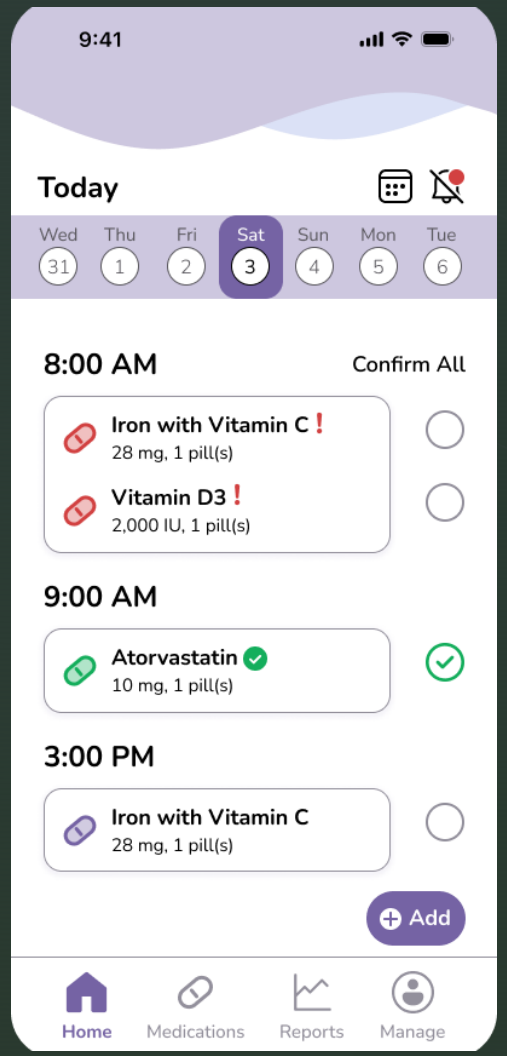


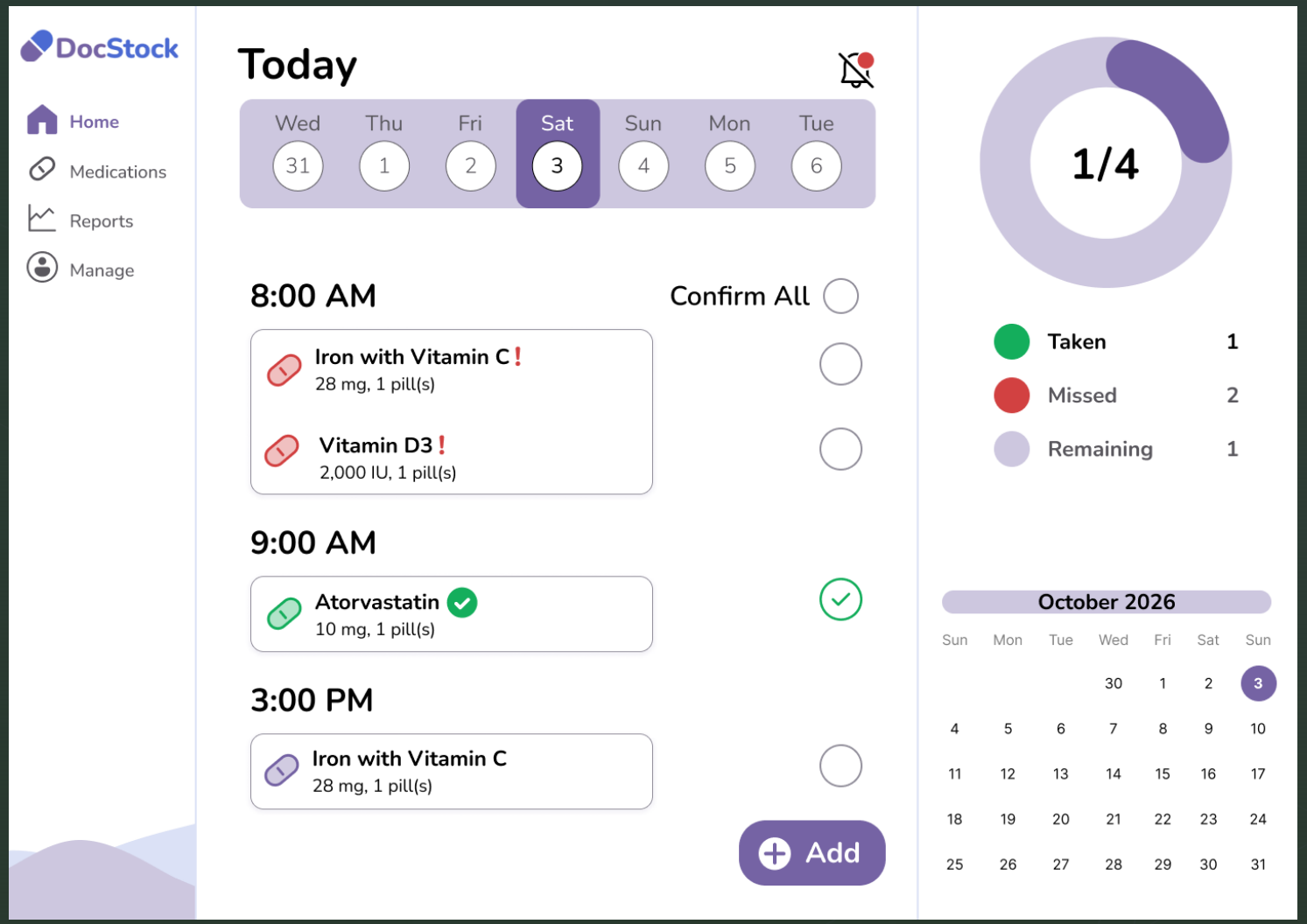
GUI Mockup: Log In Screen





GUI Mockup: Home Screen





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[11] M. Caceres and K. Rosylight, Eds., "Push API," W3C Working Draft, Dec. 1, 2025. [Online]. Available: https://www.w3.org/TR/push-api/

Appendix

User Stories:

As a user, I want to create an account so that I can manage my medications and schedules.

As a user, I want to log in securely so that only authorized people can access my medication information.

As a user, I want to specify other users as caregivers.

As a user, I want to change my caregiver's permissions so they can view or manage my medications and schedules.

As a user, I want to be able to add medications with its name, dosage, instructions, and prescribing doctor.

As a user, I want to edit medication details so I can change my prescription's information after I have added it.

Appendix Continued

User Stories Continued:

As a user, I want to view all my active medications so that I can see what I am currently taking.

As a user, I want to be able to generate an adherence report.

As a user, I want to mark medication as inactive so that medications I no longer take do not alert me.

As a user, I want to record when I take medications.

As a user, I want to create a medication schedule so I know when to take each medication.

As a user, I want to specify the frequency of a medication so I can follow a daily or other recurring schedule.

As a user, I want to set a start and end date for medications.

Appendix Continued

User Stories Continued:

As a user, I want to view upcoming medication doses so that I can plan my day with my routine in mind.

As a user, I want to update a schedule when a prescribed regiment changes.

As a user, I want to select when I have taken a medication.

As a user, I want to receive notifications to alert me to take my medication.

As a user, I want to be able to set up recurring notifications until I acknowledge one.

As a user, I want to select if I have taken a medication, want to delay a medication, or missed a medication when I receive a notification.

Appendix Continued

User Stories Continued:

As a caregiver, I want to view my care recipient's medications and schedules,

As a caregiver, I want to manage my care recipient's medications and schedules if I have the proper permissions.

As a user, I want to see my current medication supply.

As a user, I want to update the quantity remaining of my supply.

As a user, I want to set up a refill threshold so I know when to order a refill.

As a user, I want to see if any of my medications have interactions with each other, so I may plan my schedule accordingly.